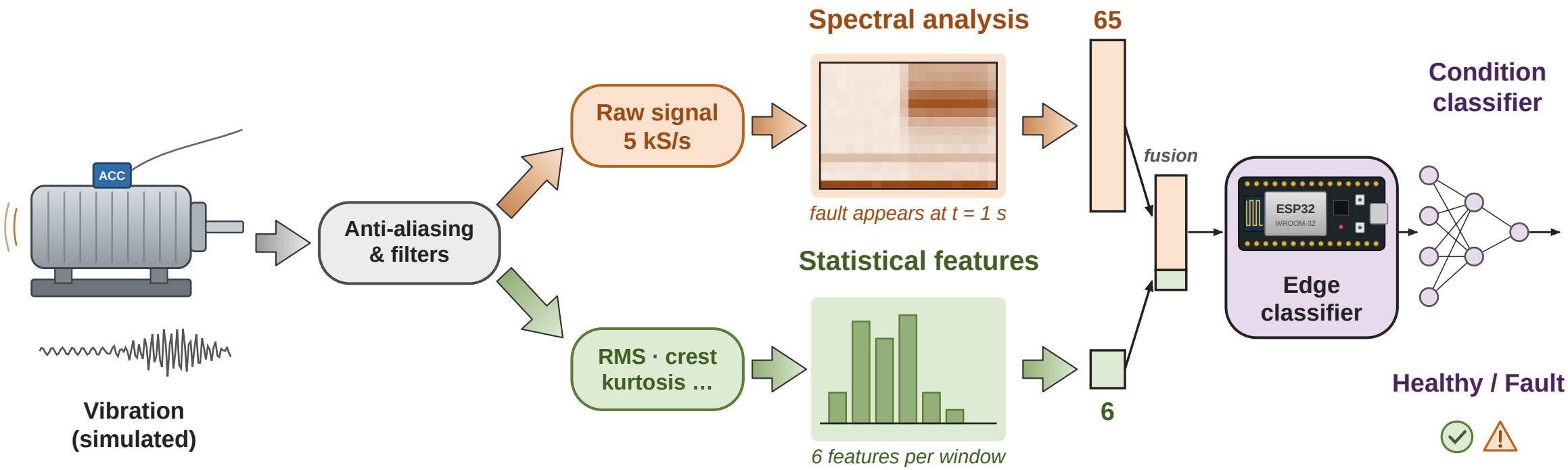Condition monitoring with an embedded classifier